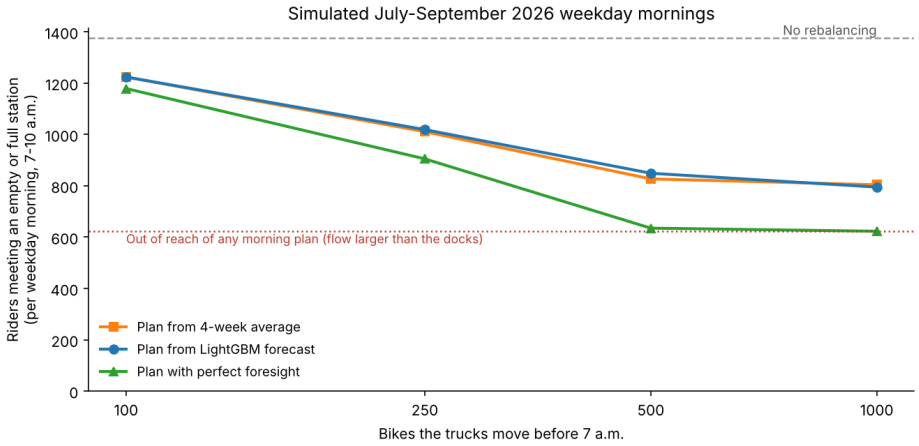
Toronto Bike Share: From 6.8 Million Trips to a Morning Rebalancing Plan

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The question. Every weekday morning, commuters ride Bike Share from residential areas into downtown. Some stations run out of bikes and others run out of free docks, so riders cannot rent or return a bike. Before 7 a.m., the operator's trucks can move a limited number of bikes. Where should they take bikes from and where should they bring them, and how much does good planning help?

What I did. I used every trip from January 2025 to September 2026 (14.6 million) in three steps. *What happened* (SQL): I cleaned the 2026 records (99.4% kept) and found where and when the system comes under strain. *What will happen* (Python): I forecast citywide demand by the hour and each station's morning rush using hourly weather, and tested the forecasts on July–September 2026, which the models never saw while being built. *What to do* (optimisation): an integer programme chooses how many bikes each station should hold at 7 a.m., with a small safety buffer against forecast error. I replayed all 63 working-weekday mornings of July–September 2026 with what riders actually did.



Riders meeting an empty or full station per weekday morning, by plan and number of bikes moved (simulation, each station starting half full).

What I found.

- **Bikes flow into the Financial District and do not come back by themselves.** On a summer weekday morning, one station at King St W and Bay St receives about 277 bikes but sends out only 4. Residential stations around Fort York and CityPlace lose 20–30 bikes each.
- **Weather moves demand.** The best forecast (LightGBM) cut the hourly error from 21% to 14% of trips compared with a simple rule of thumb. Rain in an hour cuts trips by about 39%.
- **Planning pays off.** Moving 500 bikes before 7 a.m. cut the riders meeting an empty or full station from about 1,380 to about 830–850 per morning (down 39%), or about 70% of the problems a morning plan can reach.
- **A better forecast did not make a better plan.** The LightGBM forecast was 9% more accurate at station level, but its plan did no better than one built from a four-week average. The safety buffer mattered more.

How sure can we be? The data does not record how many bikes were at each station, so every station is assumed to start half full, and only each station's net flow over the morning is used. The results compare plans with each other; they are not a count of what actually happened. Riders who found no bike left no record, so true demand is understated.

What this means for decision-makers. For the *operator*: a simple forecast plus a safety buffer captures most of the benefit of planning; in the simulation, more truck capacity (up to about 500 bikes a morning) helped far more than a more complex model. About 620 problems a morning sit at four downtown stations whose inflow is several times their dock count. They can only cope because bikes are already cleared during the rush, so mid-morning operations there matter more than any plan made before 7 a.m. For *marketing*: one busy morning route to Mimico GO Station is used mostly by non-members, a natural target for membership offers.

Full analysis, SQL and Python code: https://github.com/betterandy24/toronto-bike-share-sql